

Compound Units: Speed, Density & Pressure

Test · 26 marks total

Name: _____

Date: _____

Score: _____ / 26

Section A — Speed

A1. Distance 150 m, time 30 s. Calculate the speed.

[2 marks]

A2. Speed 25 m/s, time 8 s. Calculate the distance.

[2 marks]

A3. Distance 90 km, speed 45 km/h. Calculate the time.

[2 marks]

A4. Convert 54 km/h to m/s.

[1 mark]

Section B — Density

B1. Mass 40 g, volume 20 cm³. Calculate the density.

[2 marks]

B2. Density 2.7 g/cm³, mass 54 g. Calculate the volume.

[2 marks]

B3. Convert 1.5 g/cm³ to kg/m³.

[2 marks]

Section C — Pressure

C1. Force 300 N, area 0.5 m². Calculate the pressure in Pa.

[2 marks]

C2. Pressure 4000 Pa, area 0.25 m². Calculate the force.

[2 marks]

C3. Explain, using the pressure formula, why a sharp knife cuts more easily than a blunt one.

[2 marks]

Section D — Mixed & converting

D1. A cyclist travels 12 km in 40 minutes. Calculate their average speed in km/h, then convert your answer to m/s.

[3 marks]

D2. A block has mass 250 g and volume 100 cm³. Calculate its density in g/cm³, then convert your answer to kg/m³.

[2 marks]

D3. A force of 150 N acts on an area of 0.03 m². Calculate the pressure. State what would happen to the pressure if the area were halved and the force stayed the same.

[2 marks]